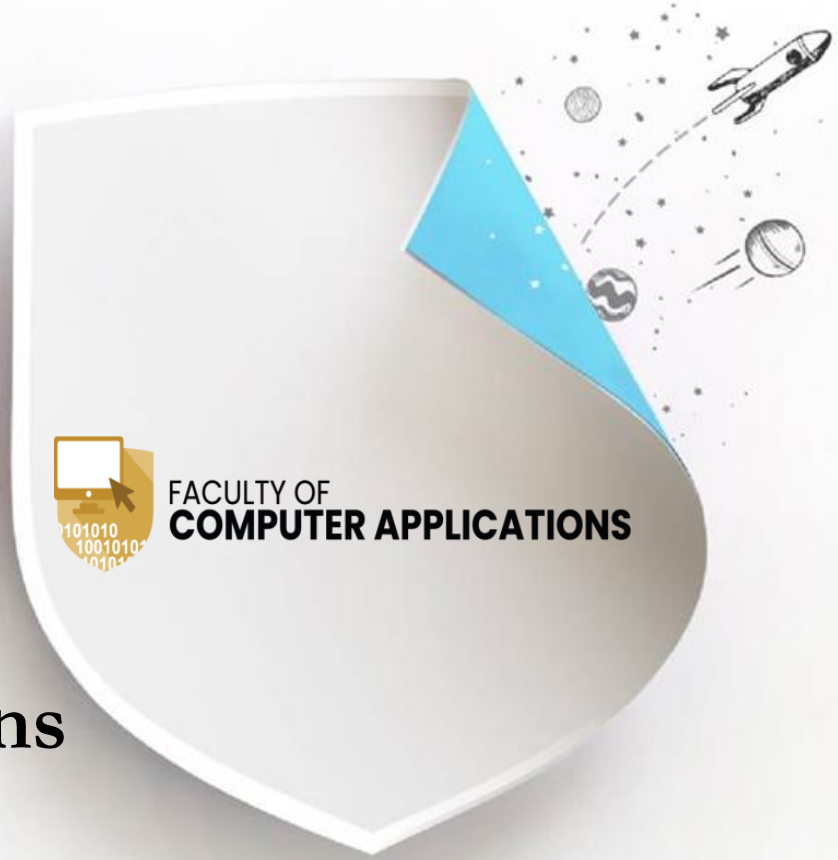
Marwadi University
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NAAC A+



FACULTY OF COMPUTER APPLICATIONS

Bachelor of Computer Applications

BCA Semester: 2

05BC3202 – Analysis and Design of Systems

PROGRAM	BACHELOR OF COMPUTER APPLICATIONS
SEMESTER	2
COURSE TITLE	ANALYSIS AND DESIGN OF SYSTEMS
COURSE CODE	05BC3202
COURSE CREDITS	4
COURSE DURATION	20 HOURS

UNIT 5 – QUALITY ASSURANCE AND IMPLEMENTATION

TOPICS COVERED

- DESIGNING ACCURATE DATA ENTRY PROCEDURES:
  - Effective coding
  - Effective and efficient data capture
  - Ensuring data quality through input validation

- QUALITY ASSURANCE AND IMPLEMENTATION:
  - TQM approach
  - Documentation approach
  - Testing, maintenance and auditing
  - Implementing distributed systems
  - Training users
  - Conversion to a new system
  - Evaluation

QUALITY ASSURANCE AND IMPLEMENTATION

- One of the ways that data can be entered more accurately and efficiently is through the knowledgeable employment of various codes.
- The process of putting ambiguous or cumbersome data into short, easily entered digits or letters is called coding (not to be confused with program coding).
- Coding aids the systems analyst in reaching the objective of efficiency, because data that are coded require less time for people to enter, and thus reduce the number of items entered.
- Coding can also help in the appropriate sorting of data at a later point in the data transformation process.
- In addition, coded data can save valuable memory and storage space.

HUMAN PURPOSES FOR CODING & OBJECTIVES OF EFFECTIVE CODING

1. Keeps track of something
2. Classifies information
3. Conceals information
4. Reveals information
5. Requests appropriate action

HUMAN PURPOSES FOR CODING & OBJECTIVES OF EFFECTIVE CODING

1. **Keeps track of something :**

Sometimes we want merely to identify a person, place, or thing just to keep track of it. The number can be either random or sequential, as described in the following subsection.

A. Simple séquence code
B. Alphabetic derivation codes

1.Keeps track of something

A. Simple Sequence Code:

- Mainly two types of Sequence codes are there. 1. Simple Sequence Code and 2. Block Sequence code (block sequence code will learn in next topic)
- A number that is assigned to something if it needs to be numbered.
- No relation to the data itself

Order #	Product	Customer
5676	Rocking Chair/with Leather	Arthur Hook, Jr.
5677	Dining Room Chair/Upholstered	Millie Monice
5678	Love Seat/Upholstered	J. & D. Pare
5679	Child's Rocking Chair/Decals	Lucinda Morely

Using a simple sequence code to indicate the sequence in which orders enter a custom furniture shop

1.Keeps track of something

A. Simple Sequence Code:

❑ Advantages:

- ✓ Eliminates the possibility of assigning the same number
- ✓ It gives users an approximation of when the order was received
- ✓ Sequence codes should be used when the order of processing requires knowledge of the sequence in which items enter the system or the order in which events unfold.

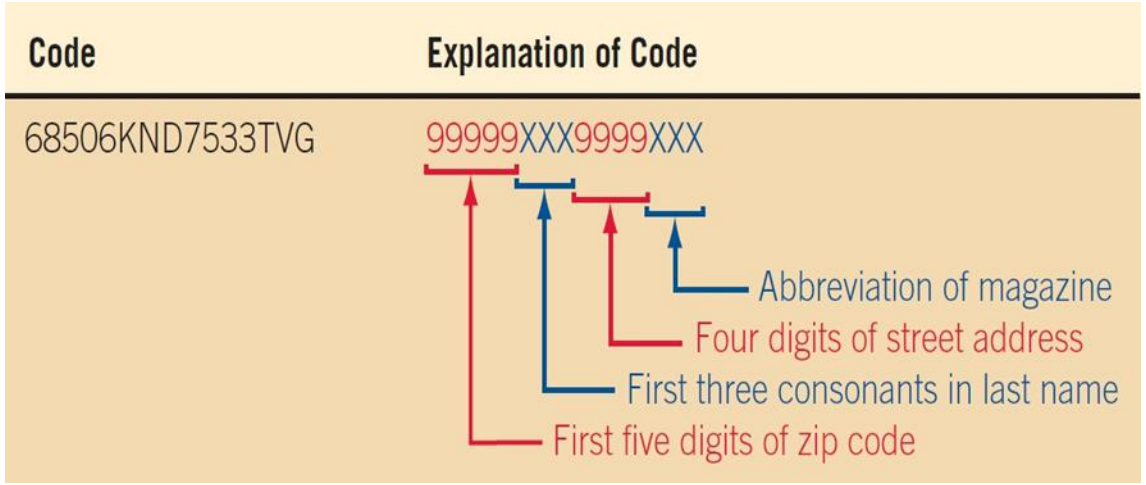
❑ Disadvantages:

- ✓ When you do not wish to have someone read the code to figure out how many numbers have been assigned
- ✓ When a more complex code is desirable to avoid a costly mistake

1.Keeps track of something

B. Alphabetic derivation codes :

- A commonly used approach in identifying an account number.



Identifying the account of a magazine subscriber with an alphabetic derivation code

2. Classifies information

Classification affords the ability to distinguish among classes of items. Classifications are necessary for many purposes, such as showing which student has completed the core requirements of his or her coursework etc.

Types:

A. Classification codes

B. Block séquence codes

2. Classifies information

A. Classification code :

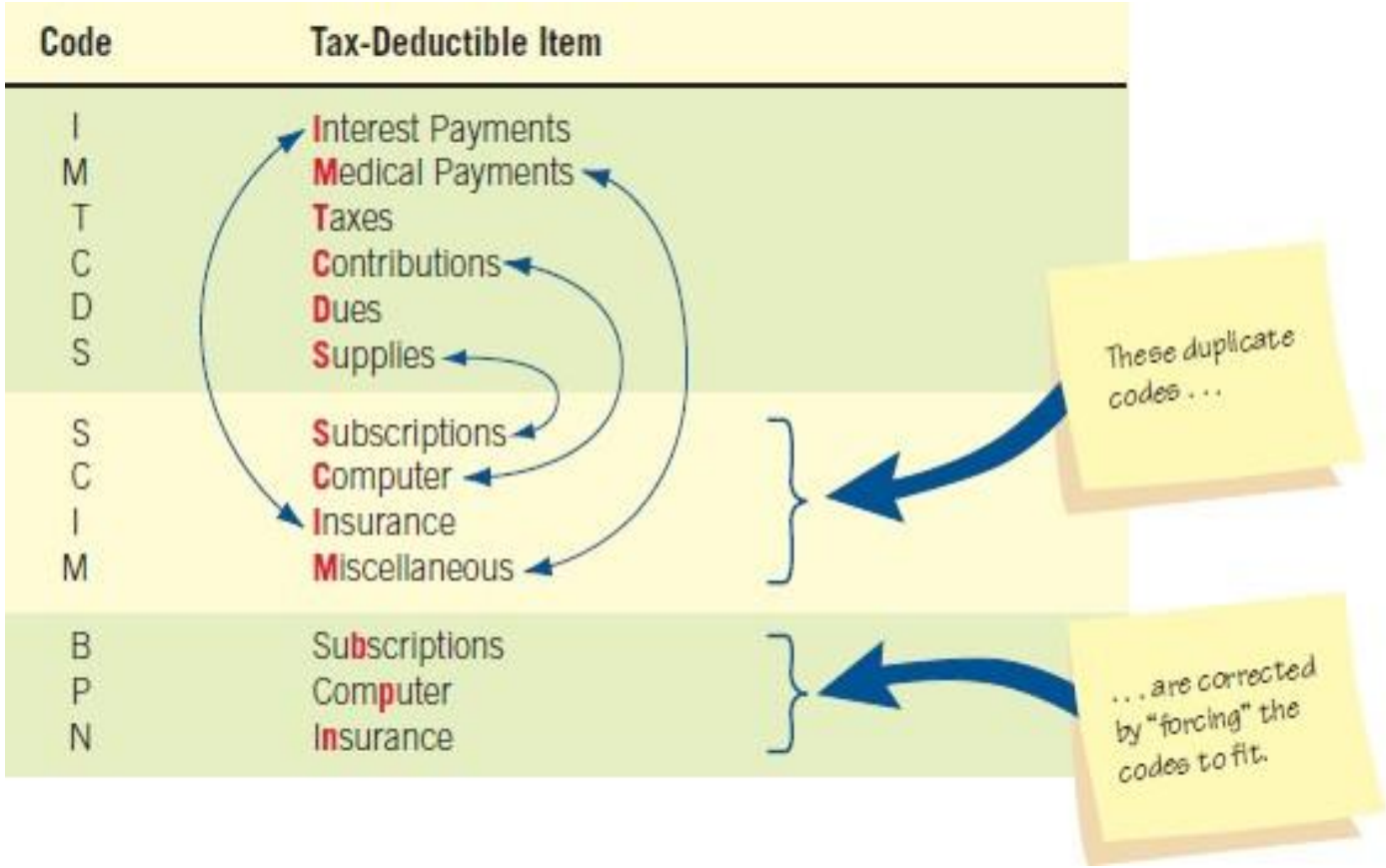
- It could include a single letter or a number and is used to separate one set of data with unique features from another.
- A concise method of characterizing an individual, location, object, or occasion
- Posted or included in manuals so that users can find them with ease.

Code	Tax-Deductible Item
I	Interest Payments
M	Medical Payments
T	Taxes
C	Contributions
D	Dues
S	Supplies

Use a single letter for a code – Example 1

2. Classifies information

A. Classification code :



Use a single letter for a code – Example 2

2. Classifies information

B. Block sequence code :

An extension of the sequence code Data are grouped according to common characteristics Simplicity of assigning the next available number (within the block) to the next item needing identification

Code	Name of Software Package	Type
100	Netscape	Browser
101	Internet Explorer	
102	Lynx	
.	.	
.	.	
.	.	
200	Access	Database
201	Paradox	
202	Oracle	
.	.	
.	.	
.	.	
300	Microsoft Word	Word Processing
301	WordPerfect	
.	.	
.	.	
.	.	
400	Astound	Presentation
401	Micrografx Designer	
402	PowerPoint	

3. Concealing Information :

- Codes may be used to conceal or disguise information we do not wish others to know. There are many reasons why a business user may want to do that.
- For example A store may want its salespeople to know the wholesale price to show them how low a price they can negotiate, but they may encode it on price tickets to prevent customers from finding that out.
- Concealing information and security have become very important in the last few years.
- Corporations have started to allow vendors and customers to access their databases directly, and handling business transactions over the Internet has made it necessary to develop tight encryption schemes.
- The following subsection describes an example of concealing information through codes.

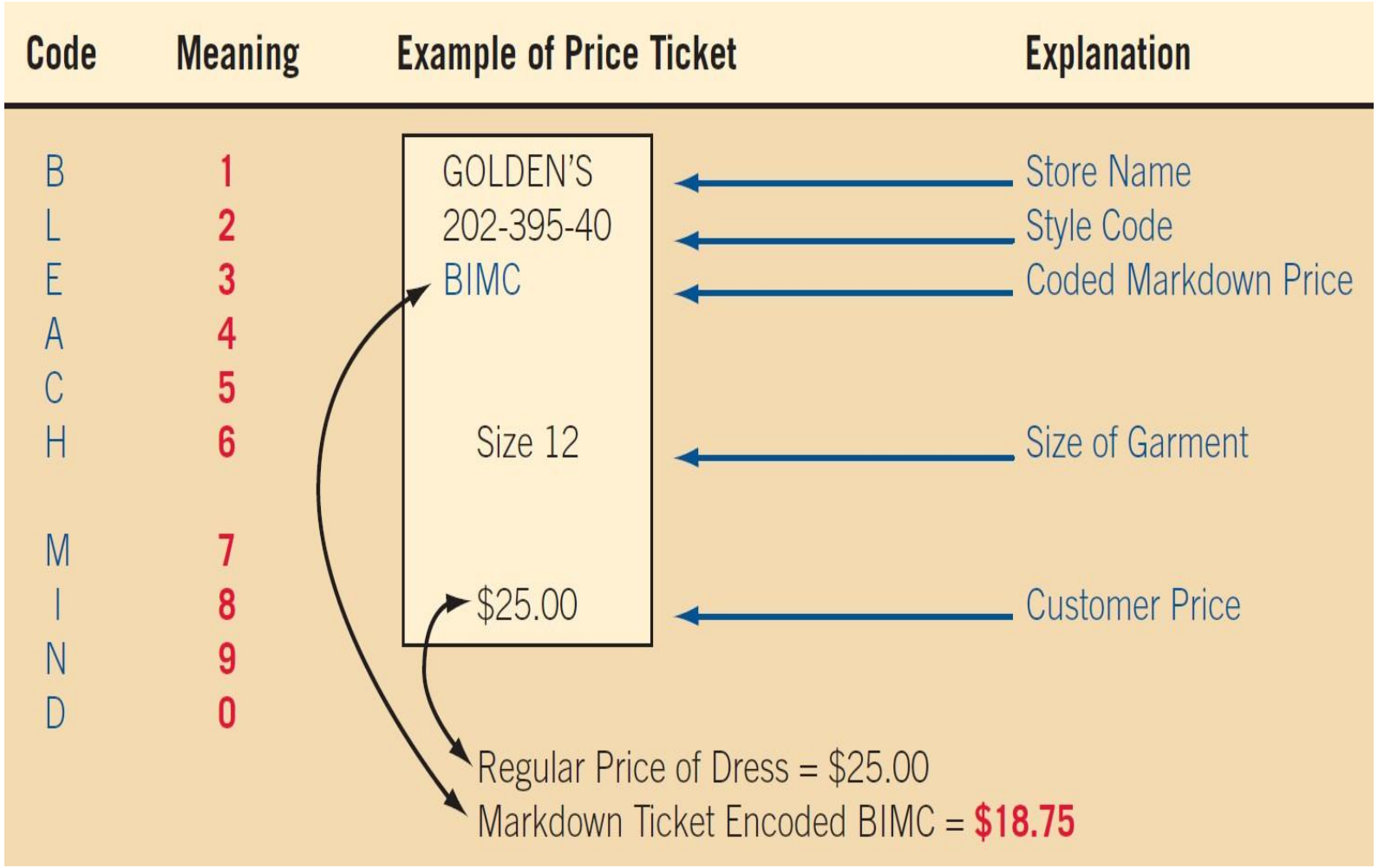
3. Concealing Information :

CIPHER CODES :

- The direct substitution of one letter for another, one number for another, or one letter for a number. A popular type of puzzle called a cryptogram is an example of letter substitution.

- Example - Encoding markdown prices with a cipher code is a way of concealing price information from customers

3. Concealing Information :



4. Reveals Information :

- Sometimes it is desirable to reveal information to specific users through a code
- Information helps the salespeople and stock people locate the place for the merchandise Makes the data entry more meaningful for humans.

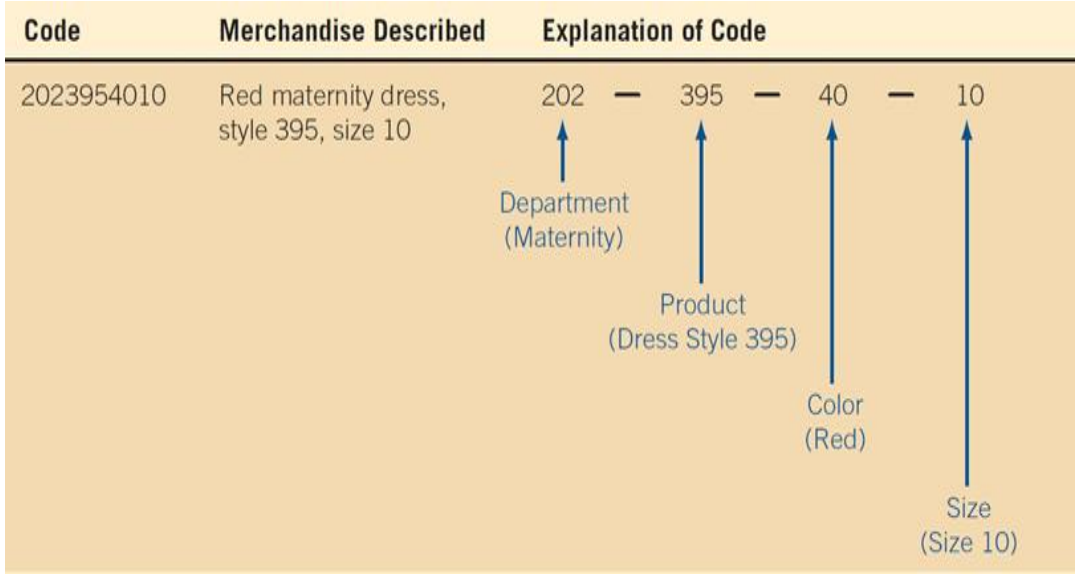
Subsections/Factors can be used in Revealing Information :

1. Significant-Digit subset codes
2. Mnemonic codes
3. Unicode

4. Reveals Information :

A. Significant-Digit subset codes :

- Used to help describe a product by virtue of its membership in many subgroups.
- Possible to locate items that belong to a certain group or class .Inquiries may be performed on a portion of the code .Useful for a marketing product.
- Significant-digit subset codes may consist of either information that actually describes the product (E.g., the number 10 means size 10), or numbers that are arbitrarily assigned (E.g., 202 is assigned to mean the maternity department)



4. Reveals Information :

B. Mnemonic codes :

- A mnemonic (pronounced nî-môn'-ĭk).
- Any code that helps the data-entry person remember how to enter the data or the user remember how to use the information is a mnemonic.
- Using a combination of letters and symbols affords a clear way to code a product so that the code is easily seen and understood.

EXAMPLE – 2
RTO codes for states in India

AN—Andaman & Nicobar Islands
AP—Andhra Pradesh
AR—Arunachal Pradesh
AS—Assam
BR—Bihar
CG—Chhattisgarh
CH—Chandigarh
DD—Daman and Diu

EXAMPLE - 1

Code	City Hospitals
BGH	**B**uffalo **G**eneral **H**ospital
ROS	**Ros**well Park Memorial Institute
KEN	**Ken**more Mercy
DEA	**Dea**coness Hospital
SIS	**Sis**ters of Charity
STF	**S**aint **F**rancis Hospital
STJ	**S**aint **J**oseph's Hospital
OLV	**O**ur **L**ady of **V**ictory Hospital

4. Reveals Information :

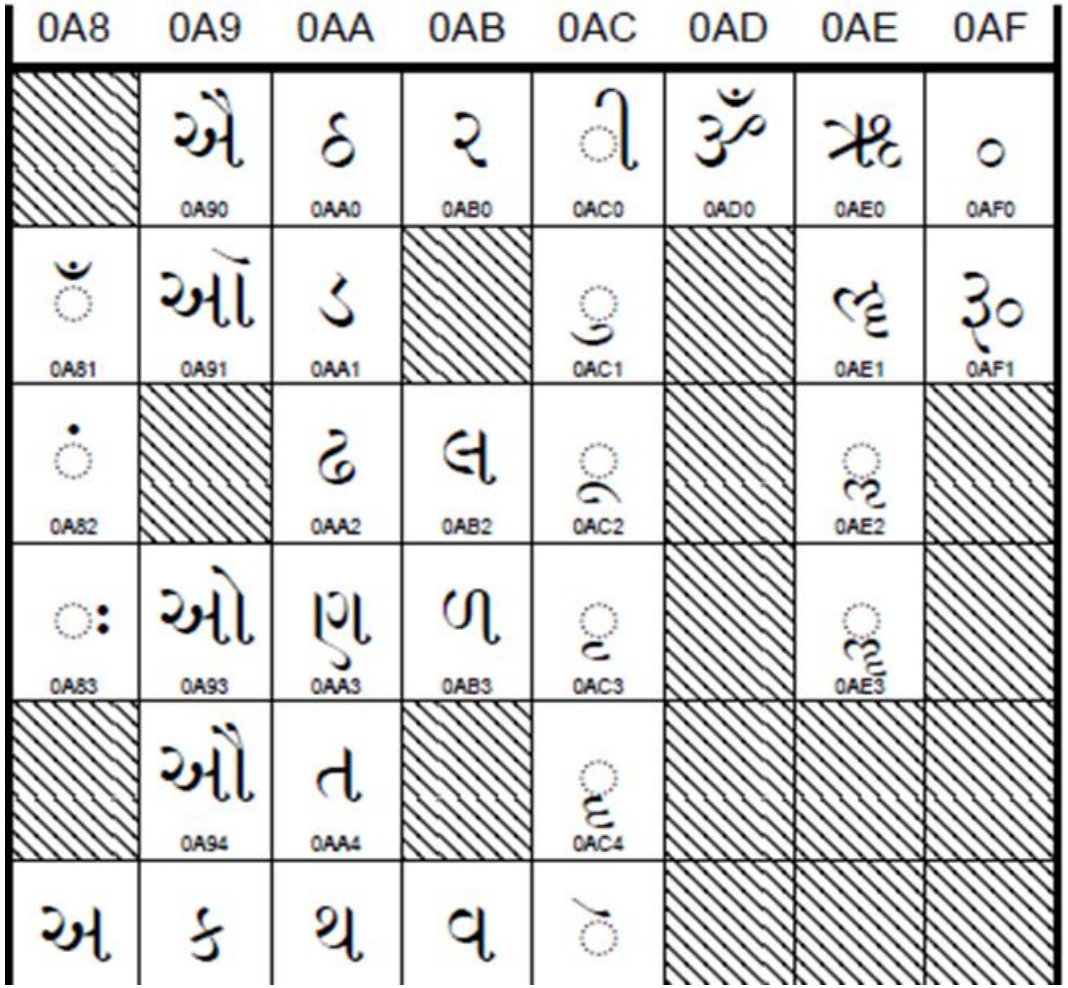
こにちわ
ko ni chi wa
hello

C. UNICODE:

- Includes all standard language symbols
- Has room for 65,535 characters
- You can display Web pages written in other alphabets by downloading an input method editor from Microsoft.
- The full set of Unicode characters are grouped by language and may be found at www.unicode.org
- Glyph symbols are represented using an "&#xnnnn;" notation, in which nnnn represents a specific letter or symbol, and x means that hexadecimal notation, or base 16 numbering, is used to represent the Unicode characters.
- The code used for the Japanese word for hello, konichiwa, is こにち&#x308F.

4. Reveals Information :

c. UNICODE:



0A8	0A9	0AA	0AB	0AC	0AD	0AE	0AF
	ઐ 0A90	ઠ 0AA0	ર 0AB0	ી 0AC0	ૐ 0AD0	ૠ 0AE0	૰ 0AF0
ઁ 0A81	ઑ 0A91	ડ 0AA1		ુ 0AC1		ૡ 0AE1	૱ 0AF1
ં 0A82		ઢ 0AA2	લ 0AB2	ૂ 0AC2		ૢ 0AE2	
ઃ 0A83	ઓ 0A93	ણ 0AA3	ળ 0AB3	ૃ 0AC3		ૣ 0AE3	
	ઔ 0A94	ત 0AA4		ૄ 0AC4			
અ	ક	થ	વ	ૅ			

5. Requesting Appropriate Action:

- Instruct either the computer or the decision maker about what action to take.
- Short numeric or alphanumeric codes used to spell out precisely what activities are to be accomplished
- Function codes

Example :

Code	Function
1	Delivered
2	Sold
3	Spoiled
4	Lost or Stolen
5	Returned
6	Transferred Out
7	Transferred In
8	Journal Entry (Add)
9	Journal Entry (Subtract)

GENERAL GUIDELINES FOR CODING

1. Be concise
2. Keep the codes stable
3. Ensure that codes are unique
4. Allow codes to be sortable
5. Avoid confusing codes
6. Keep the codes uniform
7. Allow for modification of codes
8. Make codes meaningful
9. Using CODE

GENERAL GUIDELINES FOR CODING

1. **Be concise**

- Overly long codes mean more keystrokes and consequently more errors and it also mean that storing the information in a database will require more memory.
- Short codes are easier for people to remember and easier to enter.
- If codes must be long, they should be broken up into subcodes E.g. 5678-923-453-127
- Sometimes necessary to have large number E.g. Credit card number

GENERAL GUIDELINES FOR CODING

2. Keep the codes stable :

- Stability means that the identification code for a customer should not change each time new data are received
- Don't change the codes in a mnemonic system

3. Ensure That Codes Are Unique :

- For codes to work, they must be unique Do not assign the same code number or name to the same items

GENERAL GUIDELINES FOR CODING

4. Allow Codes to Be Sortable :

- Make sure that users can do what you intend to do with the codes you create
- Numerical codes are much easier to sort than alphanumeric data

Incorrect Sorting Using MMM-DD-YYYY	Incorrect Sorting Using MM-DD-YYYY	Incorrect Sorting (Year 2000 Problem) YY-MM-DD	Correct Sorting Using YYYY-MM-DD
Dec-25-1998	06-04-1998	00-06-11	1997-06-12
Dec-31-1997	06-11-2000	97-06-12	1997-12-31
Jul-04-1999	06-12-1997	97-12-31	1998-06-04
Jun-04-1998	07-04-1999	98-06-04	1998-10-24
Jun-11-2000	10-24-1998	98-10-24	1998-12-25
Jun-12-1997	12-25-1998	98-12-25	1999-07-04
Oct-24-1998	12-31-1997	99-07-04	2000-06-11

GENERAL GUIDELINES FOR CODING

5. Avoid Confusing Codes :

- Avoid using coding characters that look or sound alike. The characters O (the letter oh) and 0 (the number zero) are easily confused, as are the letter I and the number 1, and the letter Z and the number 2. Therefore, codes such as B1C and 280Z are unsatisfactory.
- One example of a potentially confusing code is the Canadian Postal Code, as shown in figure. The code format is X9X 9X9, where X stands for a letter and 9 stands for a number.

GENERAL GUIDELINES FOR CODING

- One advantage to using letters in the code is to allow more data in a six-digit code (there are 26 letters, but only 10 numbers). Because the code is used on a regular basis by Canadians, the code makes perfectly good sense to them. To foreigners sending mail to Canada, however, it may be difficult to tell if the second-to-last symbol is a Z or a 2.
- Combining look-alike characters in codes can result in errors

Code Format for Canadian Postal Code
X9X 9X9

Handwritten Code	Actual Code	City, Province	Problem
L8S 4M4	L8S 4M4	Hamilton, Ontario	S looks like a 5
T3A ZE5	T3A 2E5	Calgary, Alberta	2 looks like a Z 5 looks like an S
L0S 1J0	L0S 1J0	Niagara-on-the-Lake, Ontario	Zero and Oh look alike S looks like a 5 1 looks like an I

GENERAL GUIDELINES FOR CODING

6. Keep the Codes Uniform :

- Need to follow readily perceived form.
- Keep codes uniform among as well as within programs.
- Codes used together, such as BUF-234 and KU-3456, are poor because the first contains three letters and three numbers, whereas the second has only two letters followed by four numbers.
- When you are required to add dates, try to avoid using the codes MMDDYYYY in one application, YYYYDDMM in a second, and MMDDYY in a third. It is important to keep codes uniform among as well as within programs.

GENERAL GUIDELINES FOR CODING

7. Allow for Modification of Codes:

- Adaptability is a key feature of a good code.
- The system will evolve over time
- The coding system should be able to encompass change

8. Make Codes Meaningful :

- Effective codes contain information should make sense to people using them
- Easier to understand, work with, and recall.
- The job of data entry becomes more interesting when working with meaningful codes instead of just entering a series of meaningless numbers.

GENERAL GUIDELINES FOR CODING

9. Using Codes :

- Validation programs - to ensure that only valid codes have been entered.
- Report and inquiry programs - - to display code meanings. Reports and displays should not print the actual codes.
- GUI programs - to create drop-down lists.

CHOOSING A DATA ENTRY METHOD

A. Keyboards
B. Optical character recognition
C. Magnetic ink character recognition
D. Mark-sense forms
E. Bar codes
F. RFID

A. Keyboards :

Special function keys to open programs. Keys used to scroll and explore the Web. Ergonomic keyboards and infrared or Bluetooth-enabled keyboards.

CHOOSING A DATA ENTRY METHOD

A. Keyboards :

Special function keys to open programs. Keys used to scroll and explore the Web. Ergonomic keyboards and infrared or Bluetooth-enabled keyboards.

CHOOSING A DATA ENTRY METHOD

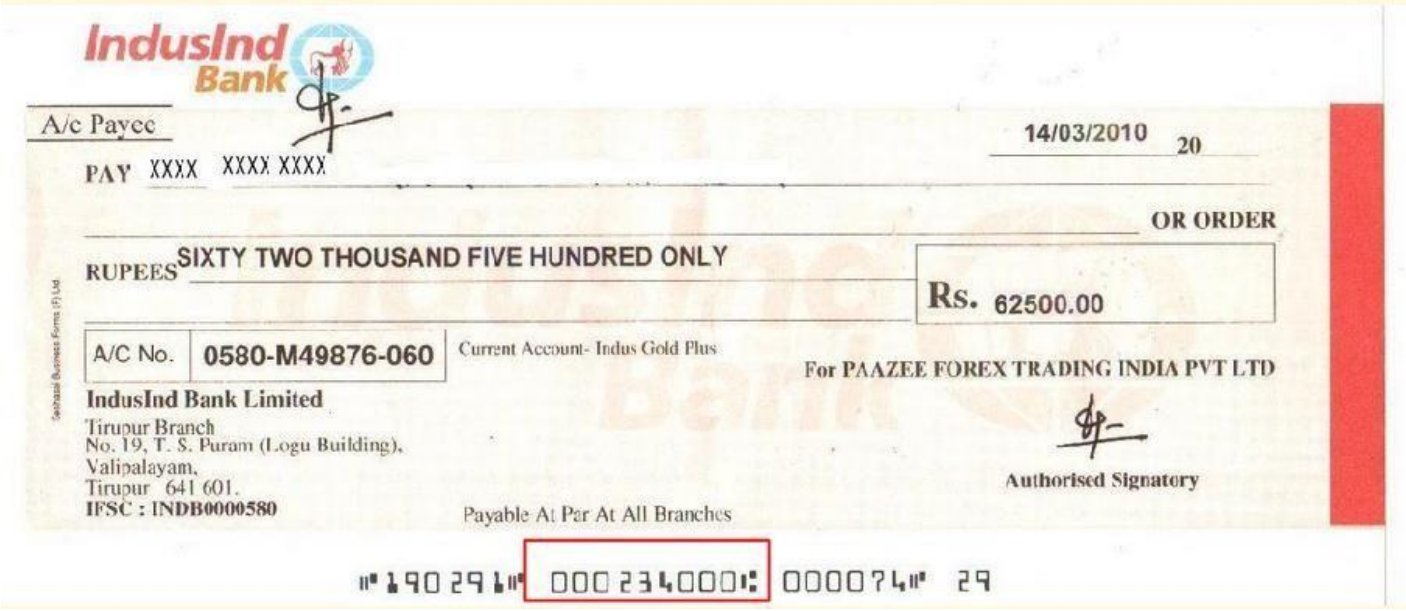
B. Optical Character Recognition :

Optical character recognition or optical character reader is the electronic or mechanical conversion of images of typed, handwritten or printed text into machine-encoded text, whether from a scanned document, a photo of a document, a scene-photo or from subtitle text on an image. OCR demands few employee skills and less training, resulting in fewer errors and less time spend by employees in redundant efforts. The transformation of faxes into documents that can be edited.

CHOOSING A DATA ENTRY METHOD

C. Magnetic Ink Character Recognition :

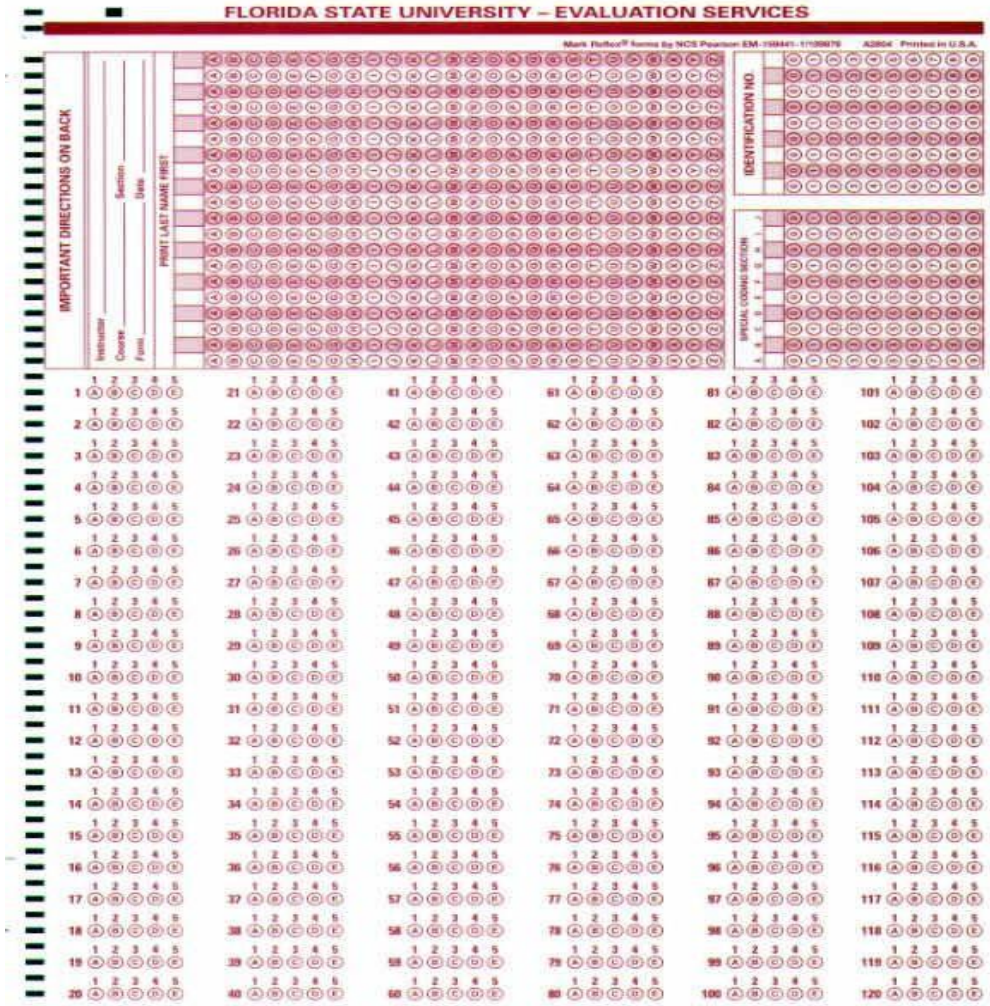
A reliable and high-speed method that is not susceptible to accepting stray marks. If it is required on all withdrawal checks, it serves as a security measure against bad checks. Data entry personnel can see the numbers making up the code if it is necessary to verify it.



CHOOSING A DATA ENTRY METHOD

D. Mark-Sense FORM:

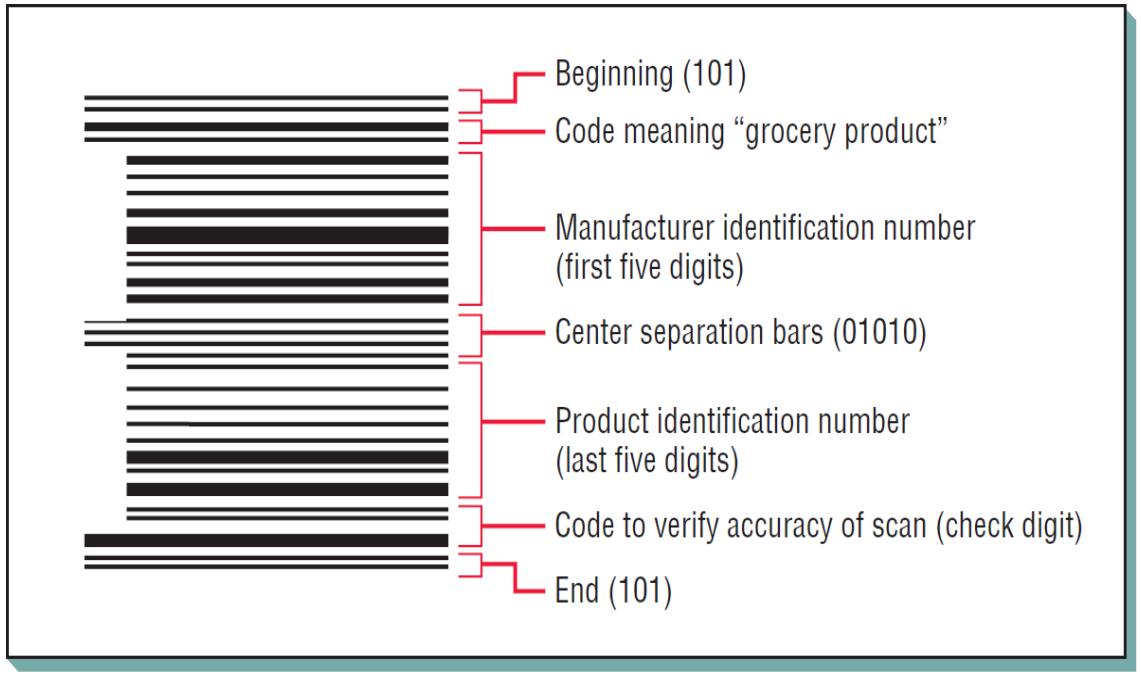
- Mark-sense forms allow data entry through the use of a scanner that senses where marks have been made on special forms.
- Little training of entry personnel is necessary
- A high volume of forms can be processed quickly
- Stray marks on form can be entered as incorrect data
- Choices are limited to the answers provided
- Difficulty in capturing alphanumeric data
- Easy to get confused and put a mark in an incorrect position



CHOOSING A DATA ENTRY METHOD

E. Bar Codes :

- Affords a high degree of accuracy for data entry
- Saves labor costs
- Allows the automatic capturing of data
- Tracking of credit card purchases



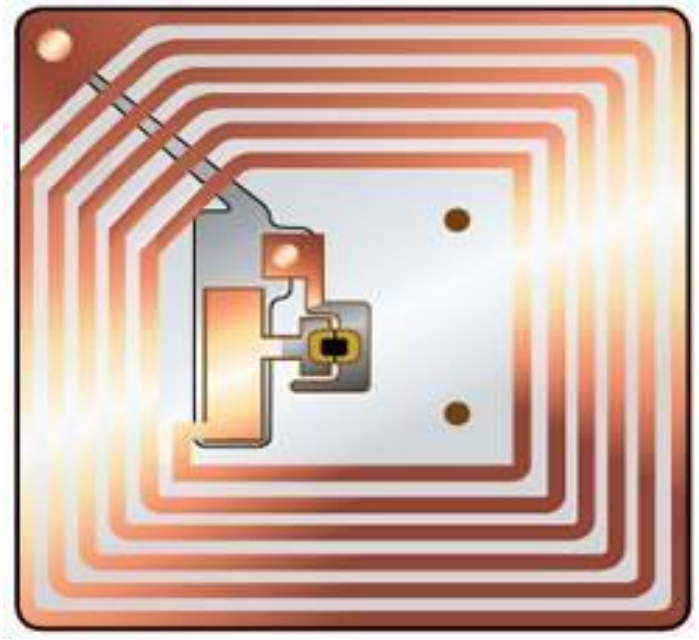
CHOOSING A DATA ENTRY METHOD

F. RFID

- Radio-frequency identification uses electromagnetic fields to automatically identify and track tags attached to objects.

Some common uses for RFID applications include:

- pet and livestock tracking
- asset tracking and equipment
- vehicle tracking etc
- It Allows the automatic collection of data using RFID tags that contain a chip and an antenna
- Privacy is a concern in this method.



ENSURING DATA QUALITY THROUGH INPUT VALIDATION

- The critical importance of catching errors during input, prior to processing and storage cannot be overemphasized
- Potential problems
- Validating input transactions
- Validating input data

ENSURING DATA QUALITY THROUGH INPUT VALIDATION

This figure Validating input is important to ensure that most potential problems with data are eliminated early

This Type of Validation	Can Prevent These Problems
Validating Input Transactions	Submitting the wrong data Data submitted by an unauthorized person Asking the system to perform an unacceptable function
Validating Input Data	Missing data Incorrect field length Data have unacceptable composition Data are out of range Data are invalid Data do not match with stored data

THE PROCESS OF VALIDATION

- Data validation is the process of ensuring that user input is clean, correct, and useful.

- It is important to validate each field until it is either valid or an error has been detected. Then a syntax test can check the length of the data entered and check for proper class and composition. Only after the syntax is correct are the semantics, or meaning, of the data validated. This includes a range, reasonable, or value test, followed by a check digit test. GUI screens help to reduce the number of human input errors when they incorporate radio buttons, check boxes, and drop-down lists. When radio buttons are used, one should be set as the default, and the only way it would be unchecked is if the user clicks a different radio button. In the case of drop-down lists, the first choice should contain a message informing the user to change the list.

THE PROCESS OF VALIDATION

Example – 1 (Validation for mobile number)

- Mobile Number validation criteria:
- The first 10 digit can contain any number between 0 to 9.
- The mobile number can have 10 digits also by including 0 at the starting.
- The mobile number can be of 12 digits also by including 91 at the starting.

1. ‘ /\+[0-9]{2}+[0-9]{10}/s ‘ or
2. '/[0-9]{10}/s’

Note : \+ Require a + (plus signal) before the number [0-9]{2} is requiring two numeric digits before the next [0-9]{10} ten digits at the end. /s Ignores whitespace and break rows.

THE PROCESS OF VALIDATION

Example – 2 (Email Input validation)

- Email validation criteria:
- An example of pattern validation used to test an email :

Address is

[A-Za-z0-9]\w{2,}@[A-Za-z0-9]{3,}\.[A-Za-z]{3}/

- The meaning of this pattern is as follows: The first letter must be any uppercase letter, lowercase letter, or number ([A-Za-z0-9]). This is followed by two or more characters that are any letter, number, or an underscore (\w{2,}).
- There must then be an @ symbol, followed by at least three letters or numbers, a period, and exactly three characters after the period.

THE PROCESS OF VALIDATION

Character Code	Meaning Used in Regular Expression Validation
\d	Any digit 0–9
\D	Any nondigit character
\w	Any letter, number, or underscore
\W	Any character other than a letter, number, or underscore
.	Matches any character
[characters]	Matches the characters in the brackets
[char-char]	Matches the range of characters
[a–z][A–Z][0–9]	Will accept any letter or digit
[^characters]	Match anything other than the characters
[^char-char]	Match anything outside the range of characters
[^a–z]	Will accept anything except lowercase letters
{n}	Match exactly n occurrences of the preceding character
{n,}	Match at least n occurrences of the character
\s	Any white space formatting character (tab, new line, return, etc.)
\S	Any nonwhite space character

QUALITY ASSURANCE & IMPLEMENTATION

Quality has long been a concern of businesses, as it should be for systems analysts in the analysis and design of information systems.

The three approaches to quality assurance through software engineering are :

(1) Securing total quality assurance by designing systems and software with a top-down, modular approach.

(2) Documenting software with appropriate tools; and

(3) Testing, maintaining, and auditing software

QUALITY ASSURANCE & IMPLEMENTATION

- The process of ensuring that the information system is operational and then allowing users to take over its operation for use and evaluation is called **implementation.**
- Implementation concerns cloud computing, and service-oriented architecture; training users and making sure that each user understands any new roles they must take on because of the new information system; choosing a conversion strategy; providing proper security, privacy, and disaster plans; and evaluating the new or modified information system

TQM (TOTAL QUALITY MANAGEMENT APPROACH)

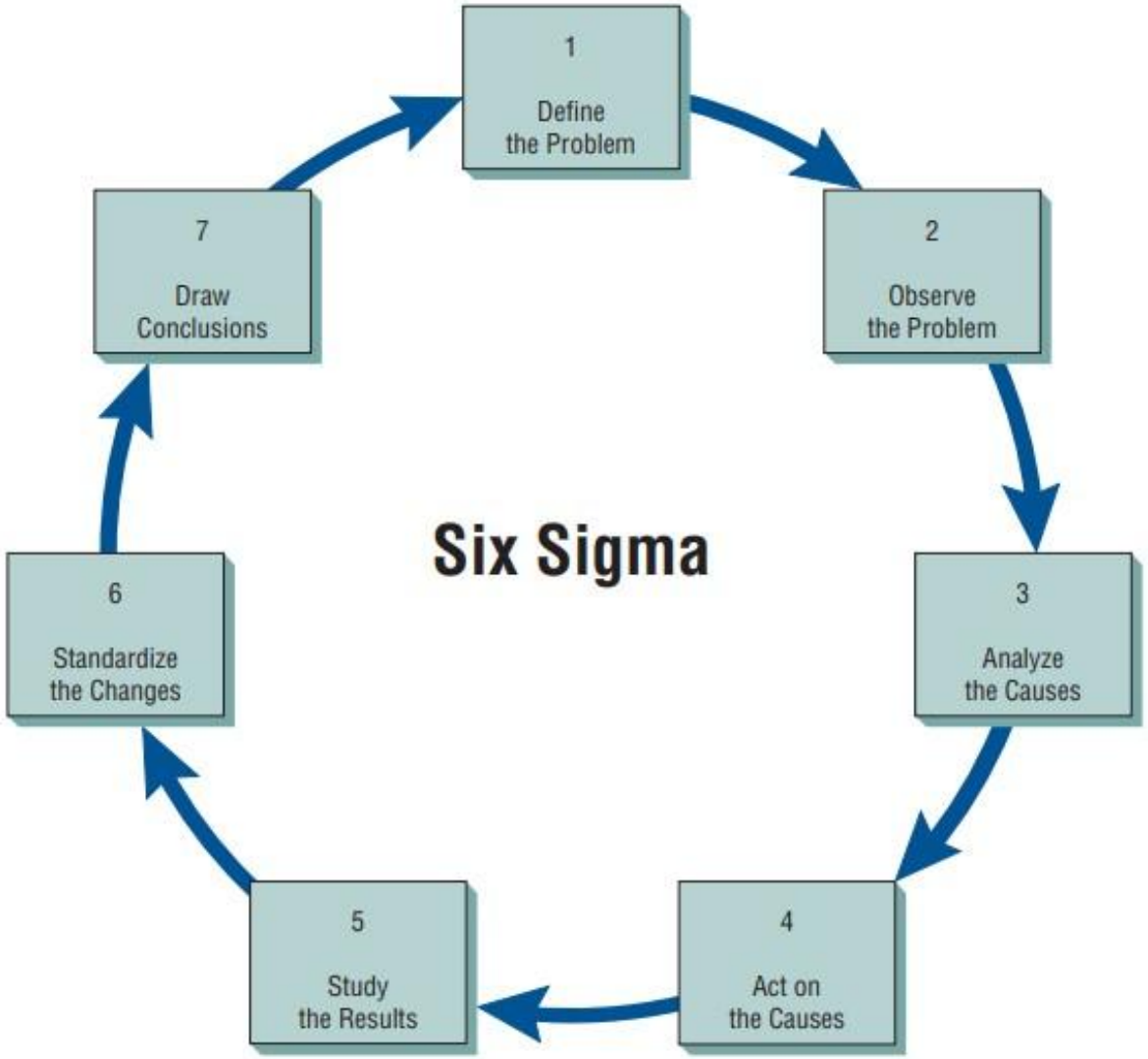
- Total quality management (TQM) is essential throughout all the systems development steps.
- According to Evans and Lindsay (2004), the primary elements of TQM are meaningful only when occurring in an organizational context that supports a quality effort.
- It is in this context that the elements of customer focus, strategic planning and leadership, continuous improvement, empowerment, and teamwork are united to change employees' behaviour and, ultimately, the organization's course.
- Notice that the concept of quality has broadened over the years to reflect an organizational, rather than an exclusively production, approach. Systems analysts must be aware of the factors that are driving the interest in quality.
- It is important to realize that the increasing commitment of businesses to TQM fits extraordinarily well into the overall objectives for systems analysis and design.

TQM (TOTAL QUALITY MANAGEMENT APPROACH)

1. Six Sigma :

- Six sigma is an approach for quality management.
- Systems analysts and systems users need to be aware of Six Sigma and apply some of the principles to their systems analysis projects. Originally developed by Motorola in the 1980.
- The goal of Six Sigma is to eliminate all defects. This applies to any product, service, or process.
- Six Sigma is a **top-down approach.**
- A Six Sigma project leader is called **a Black Belt.**
- Other project members are called **Green Belts.**
- Every systems analyst should understand the methodology and philosophy of Six Sigma which is shown in figure.

TQM (TOTAL QUALITY MANAGEMENT APPROACH)



TQM (TOTAL QUALITY MANAGEMENT APPROACH)

2. Structured Walkthrough :

- One of the strongest quality management actions the systems analysis team can take is to do structured walkthroughs routinely. Structured walkthroughs are a way of using peer reviewers to monitor the system's programming and overall development, point out problems, and allow the programmer or analyst responsible for that portion of the system to make suitable changes.

- Structured walkthroughs involve at least four people: the person responsible for the part of the system or subsystem being reviewed (a programmer or analyst), a walkthrough coordinator, a programmer or analyst peer, and a peer who takes notes about suggestions..

TQM (TOTAL QUALITY MANAGEMENT APPROACH)

3. Top Down system Designs and development:

- Many companies first introduced computer systems on the lowest level of the organization. This is where the immediate benefits to computerization are most observable and cost-effective.

- Top-down design allows the systems analyst to ascertain overall organizational objectives first, as well as to ascertain how they are best met in an overall system. Then the analyst divides that system into subsystems and their requirements.

TQM (TOTAL QUALITY MANAGEMENT APPROACH)

4. Using Structure Charts to Design Modular System/Modular Programming :

- Once the top-down design approach is taken, the modular approach is useful in programming. This approach involves breaking the programming into logical, manageable portions, or modules. Modular program design has **three main advantages.**

1. Modules are easier to write and debug.
2. Modules are easier to maintain.
3. Modules are easier to grasp, (they are self-contained subsystems)

TQM (TOTAL QUALITY MANAGEMENT APPROACH)

Some guidelines for modular programming :

1. Keep each module to a manageable size (ideally including only one function).
2. Pay particular attention to the critical interfaces (the data and control variables that are passed to other modules).
3. Minimize the number of modules the user must modify when making changes.
4. Maintain the hierarchical relationships set up in the top-down phases

TQM (TOTAL QUALITY MANAGEMENT APPROACH)

5. Service-Oriented Architecture (SOA) :

- Instead of being hierarchical like the top-down approach found in structure charts, the SOA approach is to make individual SOA services that are unassociated or only loosely coupled to one another. We can say that service-oriented architecture is simply a group of services that can be called upon to provide specific functions. Rather than including calls to other services, a service can use certain defined protocols so that it can communicate with other services.
- Each service executes one action. One service may return the number of days in this month; another may tell us if this is a leap year; a third service may reserve five nights in a hotel room from the end of February to the beginning of March. Although the third service needs to know the values obtained from the first and second services, they are independent of one another.

TQM (TOTAL QUALITY MANAGEMENT APPROACH)

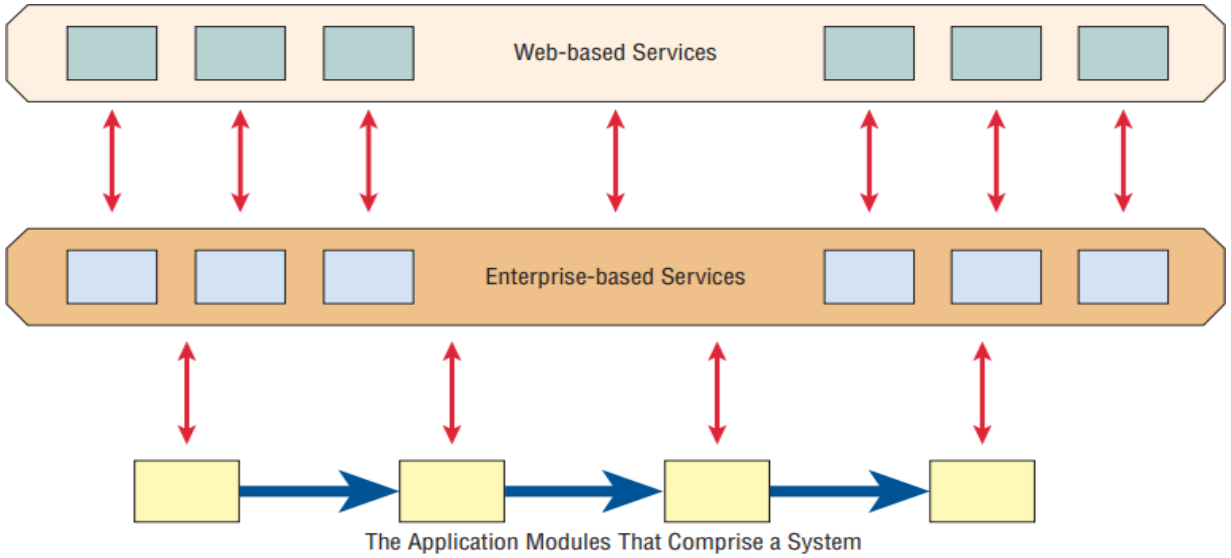
5. Service-Oriented Architecture (SOA) :

- In order to set up an SOA, the services must be:
  1. modular,
  2. reusable,
  3. work together with other modules (interoperability),
  4. able to be categorized and identified,
  5. able to be monitored, and
  6. comply with industry-specific standards

TQM (TOTAL QUALITY MANAGEMENT APPROACH)

5. Service-Oriented Architecture (SOA) :

- While the advantages of reusability and interoperability are obvious, SOA is not without its challenges. First industry standards must be agreed upon. Next, a library must be maintained so that developers can find the services they need. Finally, security and privacy can be issues when using software developed by someone else. Advocates of SOA claim that service-oriented architecture has made many of the features found in Web 2.0 possible



DOCUMENTATION APPROACH

- The total quality assurance effort requires that programs be documented properly. Software, systems, and formal and informal procedures need to be documented so that systems can be maintained and improved. Documentation allows users, programmers, and analysts to "see" the system, its software, and procedures without having to interact with it.

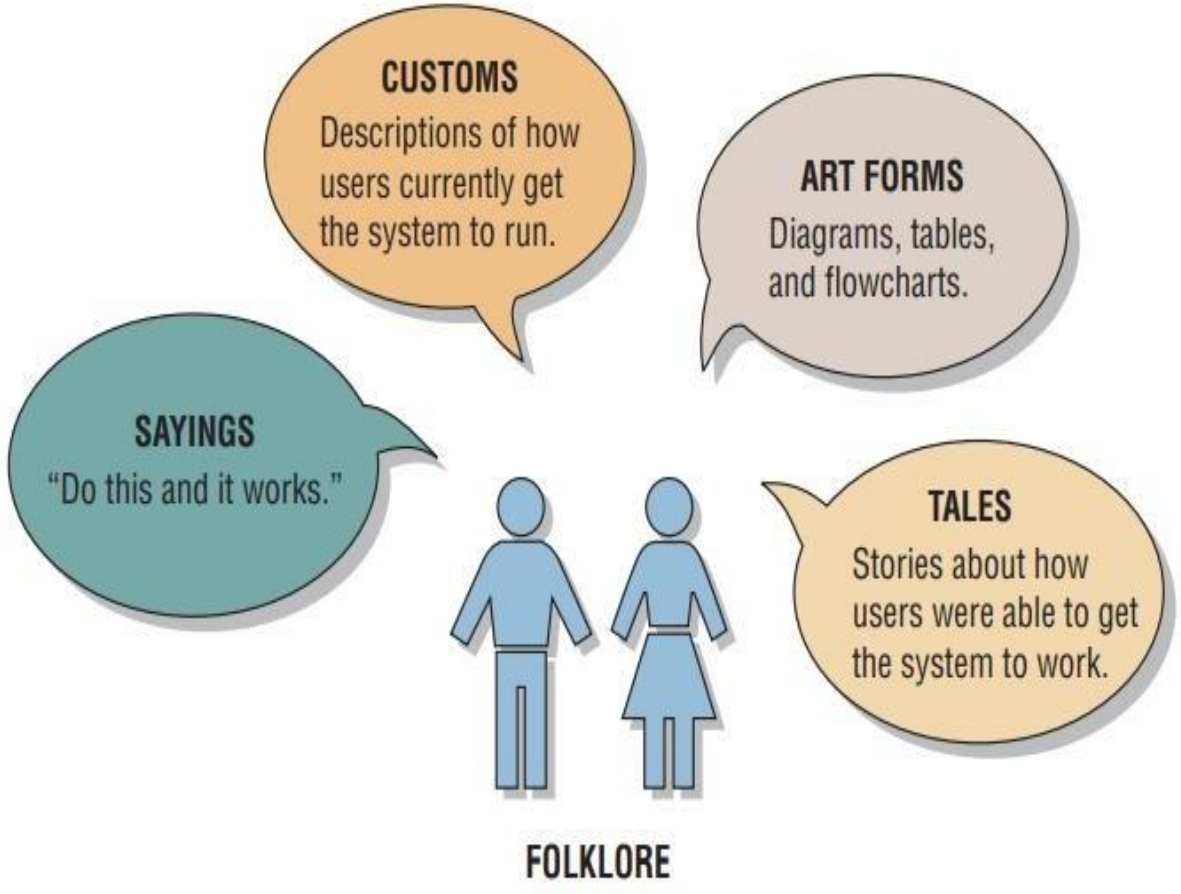
- **Procedural Manual :** Procedure manuals are common organizational documents that most people have seen. They are the English- language component of documentation, although they may also contain program codes, flowcharts, and so on

- **FOLKLORE method :** FOLKLORE method gathers information that is often shared among users but is seldom written down. This approach to systems documentation requires the analyst to interview users, investigate existing documentation in files, and observe the processing of information.

DOCUMENTATION APPROACH

- **The objective is to gather information corresponding to one of four categories: customs, tales, sayings, and art forms.**
- When documenting customs, the analyst (or other folklorist) tries to capture in writing what users are currently doing to get all programs to run without problems. An example of a custom is: "Usually, we take two days to update the monthly records because the task is quite large. We run commercial accounts on day one and save the others for the next day
- **Tales** are stories that users tell regarding how the system worked.
- **Sayings** are brief statements representing generalizations or advice
- **Gathering art** forms is another important activity of traditional folklorists, and the systems analyst should understand its importance, too. Flowcharts, diagrams, and tables that users draw

DOCUMENTATION APPROACH



TESTING, MAINTENANCE AND AUDITING

- Once the analyst has designed and coded the system, testing, maintenance, and auditing of it are prime considerations.

1. TESTING PROCESS / TESTING STEPS :

- Testing is done on many different levels at various intervals. Before the system is put into production, all programs must be desk checked, checked with test data, and checked to see if the modules work together with one another as planned.
- The system as a working whole must also be tested. Included here are testing the interfaces between subsystems, the correctness of output, and the usefulness and understandability of systems documentation and output.
- Programmers, analysts, operators, and users all play different roles in the various aspects of testing.

TESTING, MAINTENANCE AND AUDITING

STEPS FOR TESTING:

STEP 1 : PROGRAM TESTING WITH TEST DATA.

STEP 2 : LINK TESTING WITH TEST DATA.

STEP 3 : FULL SYSTEMS TESTING WITH TEST DATA.

STEP 4 : FULL SYSTEMS TESTING WITH LIVE DATA.

TESTING, MAINTENANCE AND AUDITING

STEP 1 : PROGRAM TESTING WITH TEST DATA :

At this stage, programmers must first desk check their programs to verify the way the system will work. In desk checking, the programmer follows each step in the program on paper to check whether the routine works as it is written. Throughout this process, the systems analyst checks output for errors, advising the programmer of any needed corrections. The analyst will usually not recommend or create test data for program testing but might point out to the programmer omissions of data types to be added in later tests.

STEP 2 : LINK TESTING WITH TEST DATA :

Link testing checks to see if programs that are interdependent actually work together as planned. The analyst creates special test data that cover a variety of processing situations for link testing.

TESTING, MAINTENANCE AND AUDITING

STEP 3 : FULL SYSTEMS TESTING WITH TEST DATA :

When link tests are satisfactorily concluded, the system as a complete entity must be tested. At this stage, operators and end users become actively involved in testing. Test data, created by the systems analysis team for the express purpose of testing system objectives, are used.

TESTING, MAINTENANCE AND AUDITING

STEP 4 : FULL SYSTEMS TESTING WITH LIVE DATA :

When systems tests using test data prove satisfactory, it is a good idea to try the new system with several passes on what is called live data, data that have been successfully processed through the existing system. Procedure manuals, just like computer software, also need to be tested. Although manuals can be proofread by support staff and checked for technical accuracy by the systems analysis team, the only real way to test them is to have users and operators try them, preferably during full systems testing with live data. Consider user suggestions, and incorporate them into the final versions of Web pages, printed manuals, and other documentation.

TESTING, MAINTENANCE AND AUDITING

Maintenance:

- Your objective as a systems analyst should be to install or modify systems that have a reasonably useful life. Maintenance is performed most often to improve the existing software
- Maintenance is also done to update software in response to the changing organization.

Auditing:

- Broadly defined, auditing refers to having an expert who is not involved in setting up or using a system examine information.
- For information systems, there are generally two kinds of auditors/methods: internal and external.

TESTING, MAINTENANCE AND AUDITING

Internal :

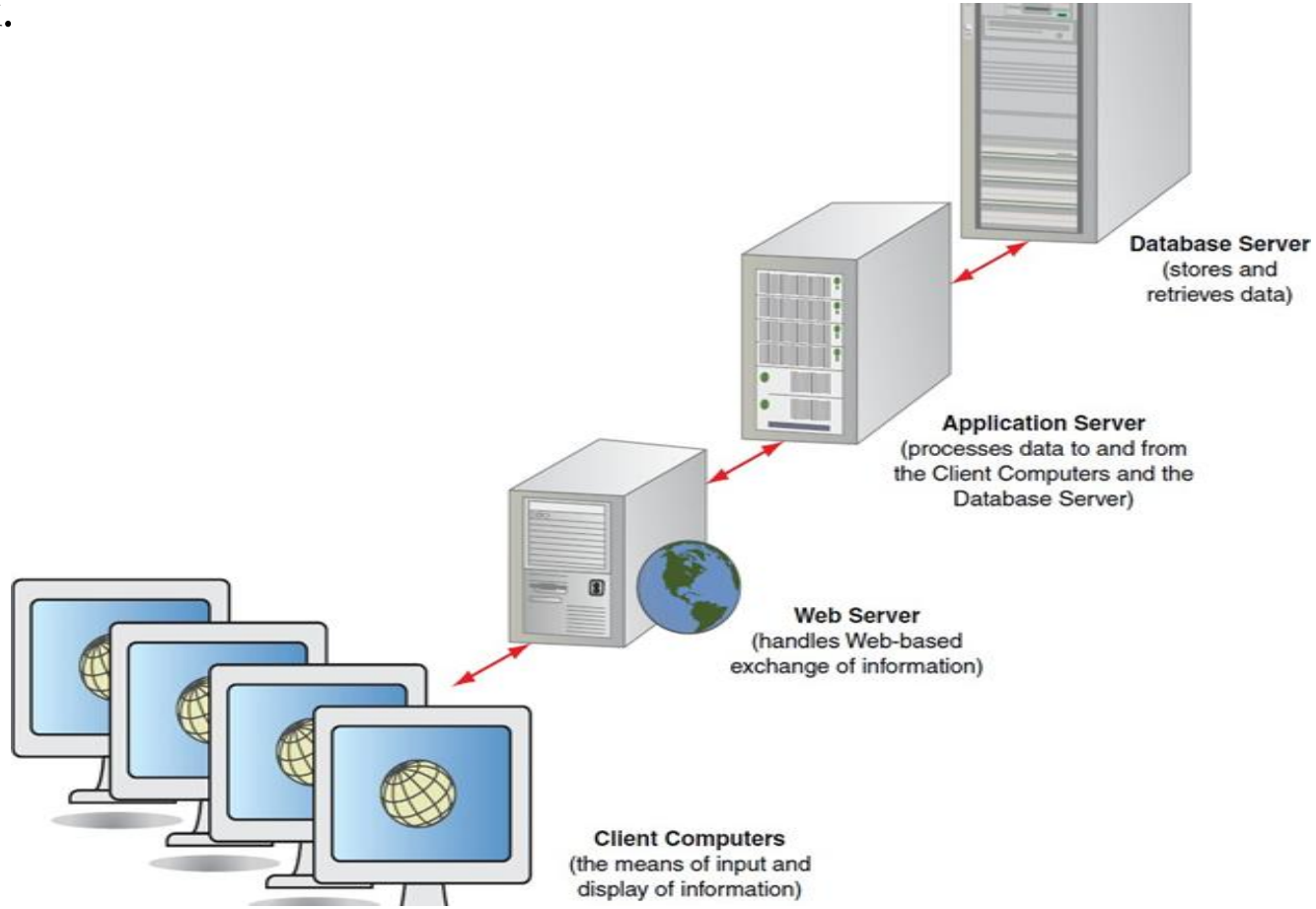
- Internal auditors study the controls used in the information
- system to make sure that they are adequate and that they are doing what they are purported to be doing. They also test the adequacy of security controls.

External :

- External auditors audit the system to ensure the fairness of the financial statements being produced. They may also be brought in if there is something out of the ordinary occurring that involves company employees, such as suspected computer fraud

IMPLEMENTING DISTRIBUTED SYSTEMS

- In this model, the processing functions are delegated either to clients (users) or to servers, depending on which machines are most suitable for executing the work.



IMPLEMENTING DISTRIBUTED SYSTEMS

Client-Server Technology :

- The client-server model, client-server computing, client-server technology, and client-server architecture all refer to a design model that can be thought of as applications running on a network.
- In very basic terms, you can picture the client requesting—and the server executing or in some way fulfilling—the request. That would be considered two-tier client-server architecture.
- When you see the term client, you might be tempted to think of people or users; for example, we speak of "clients of our consulting practice."
- In the client-server model, however, the term client refers not to people but to networked machines that are typical points of entry to the client-server system that is used by humans.
- Therefore, clients could be networked desktop computers, a workstation, or notebook computers, or any other way in which the user can enter the system.

PROS AND CONS OF CLIENT SERVER MODEL

PROS:

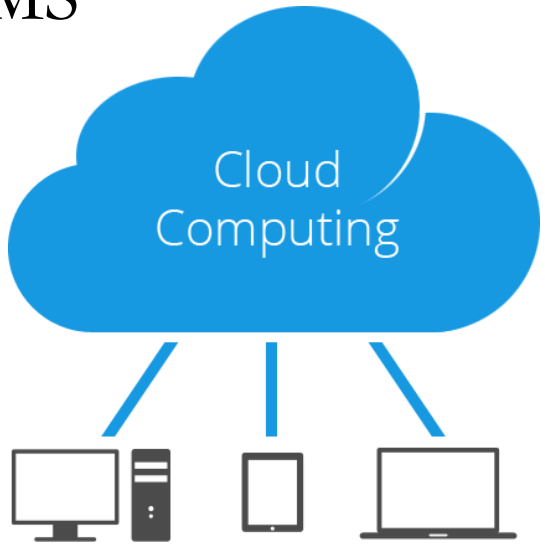
- Client server architecture has fully right to control all activities of entire network centralized.
- All data is saved on the central area.
- All network devices can be handled centrally.
- All concern like as Backups and network protection also can be managed centrally.
- All users also have right to access entire shard files which are stored centrally.

PROS AND CONS OF CLIENT SERVER MODEL

CONS:

- If, main server gets halt then entire system will be failed.
- Client server architecture is need special network operating system.
- More expensive to configure their hardware and software components
- To need well qualified technical staff for maintaining the server.
- Traffic Congestion Problem – If large number of client try to send requests at the same time frame then "Traffic Congestion Problem" can be created.
- Its cost is not affordable for normal users.

IMPLEMENTING DISTRIBUTED SYSTEMS



Cloud Computing:

- Cloud computing is the on-demand availability of computer system resources, especially data storage and computing power, without direct active management by the user.
- Large clouds often have functions distributed over multiple locations, each location being a data centre. , cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale.

NETWORKING MODELLING

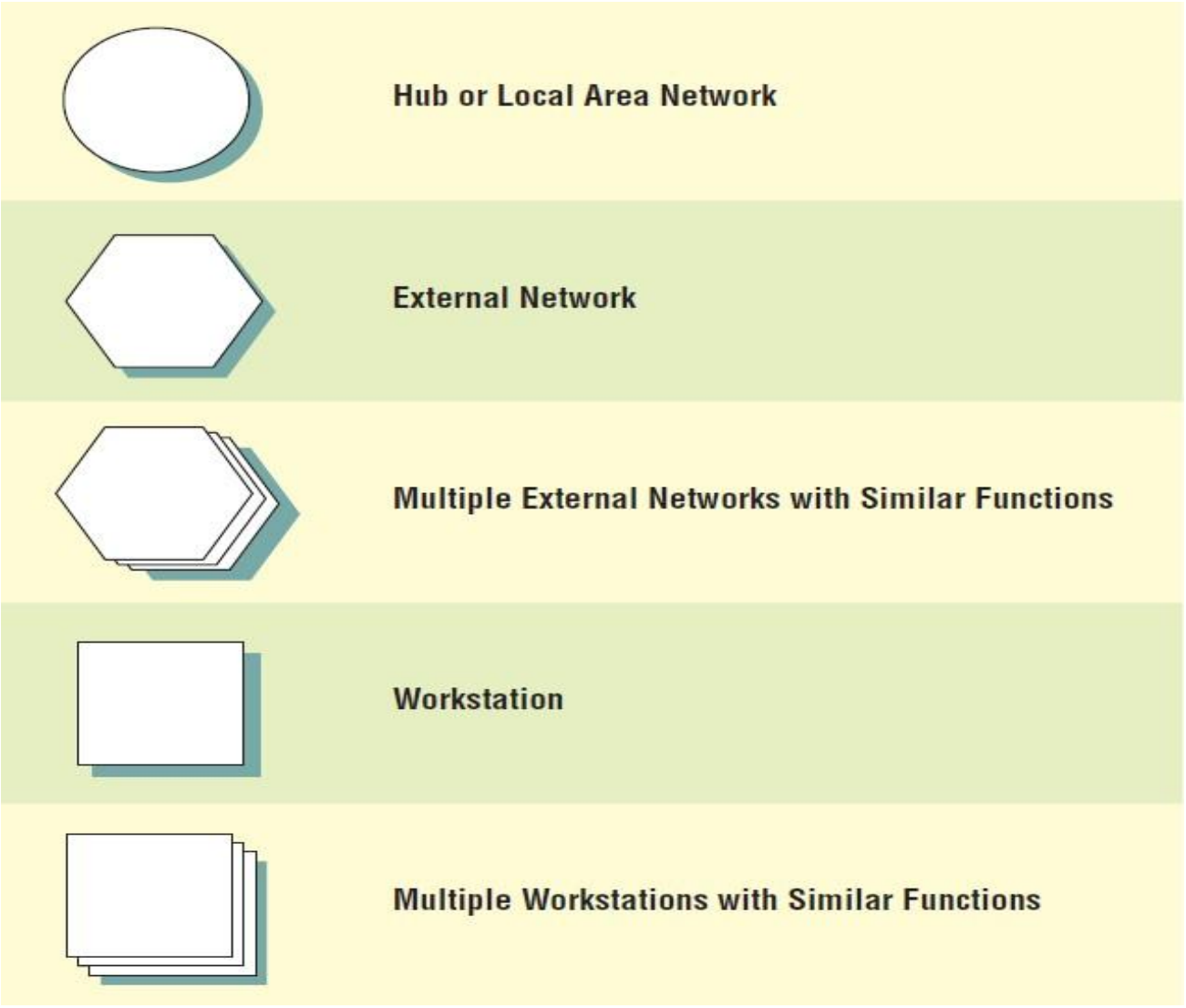
Network Modelling :

- Because networking has become so important, the systems designer needs to consider network design.

DRAWING A NETWORK DECOMPOSITION DIAGRAM :

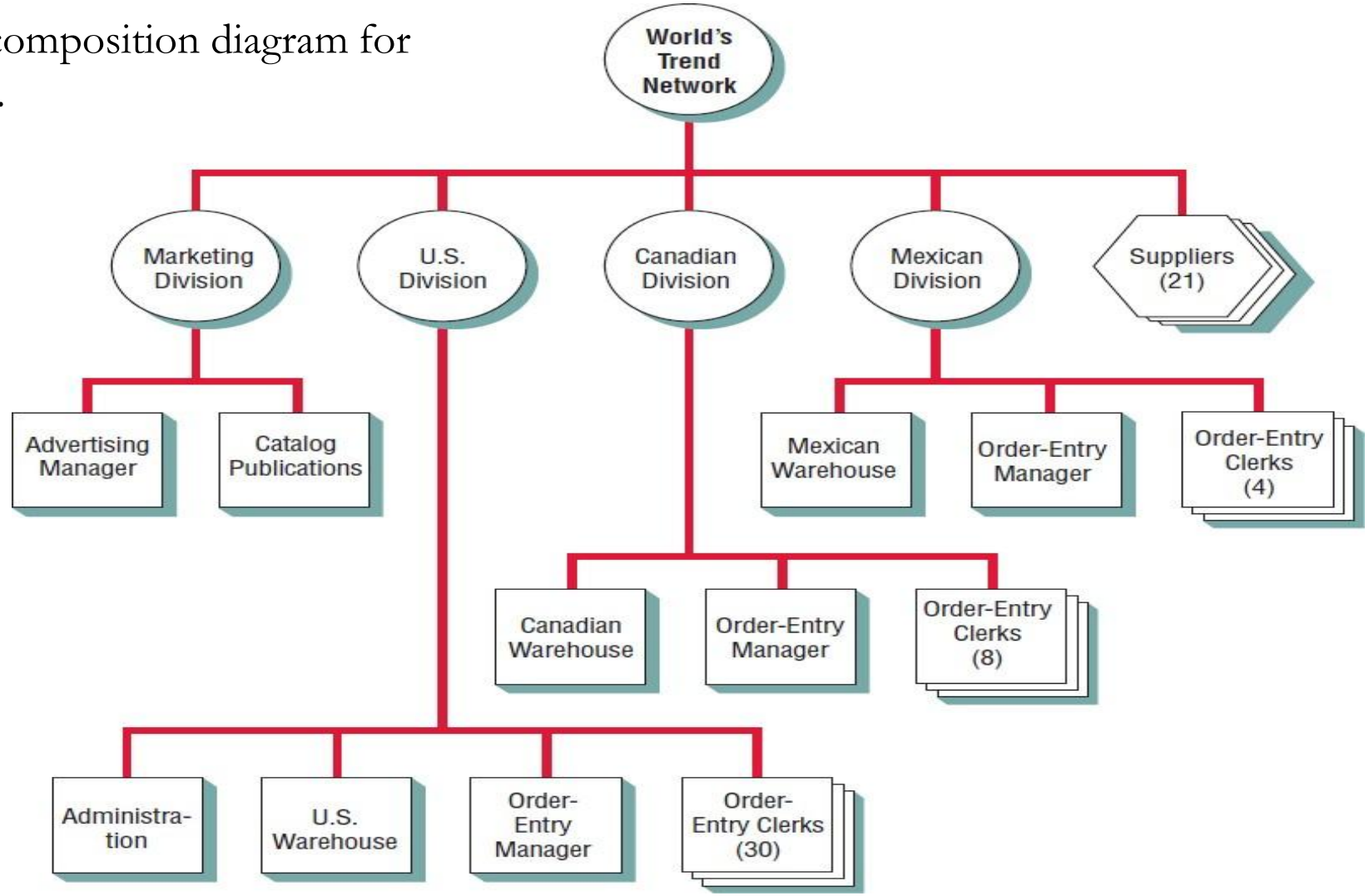
- We can illustrate drawing a network decomposition model by referring once again to the World's Trend Catalog Division example, Start by drawing a circle at the top and labelling it "World's Trend Network."
- Now draw a number of circles on the level below, These circles represent hubs for the Marketing Division and each of the three order-entry and distribution centers (U.S. Division, Canadian Division, Mexican Division).
- We can extend this drawing further by drawing another level. This time, we can add the workstations. For example, the Marketing Division has two workstations connected to it, whereas the U.S. Division has 33 workstations on its LAN (Administration, the Warehouse, the Order-Entry.

NETWORKING MODELLING



NETWORKING MODELLING

A network decomposition diagram for World's Trend.



TRAINING STRATEGIES

- Training strategies are determined by who is being trained and who will train them.
- The analyst will want to ensure that anyone whose work is affected by the new information system is properly trained by the appropriate trainer.

PEOPLE WHO TRAIN USERS :

For a large project, many different trainers may be used depending on how many users must be trained and who they are. Possible training sources include the following:

1. Vendors.
2. Systems analysts.
3. External paid trainers.
4. In-house trainers.
5. Other system users.

TRAINING STRATEGIES

❑ Guidelines for Training :

The analyst has four major guidelines for setting up training.
They are:
(1) establishing measurable objectives
(2) using appropriate training methods,
(3) selecting suitable training sites, and
(4) employing understandable training materials.

❑ Training methods :

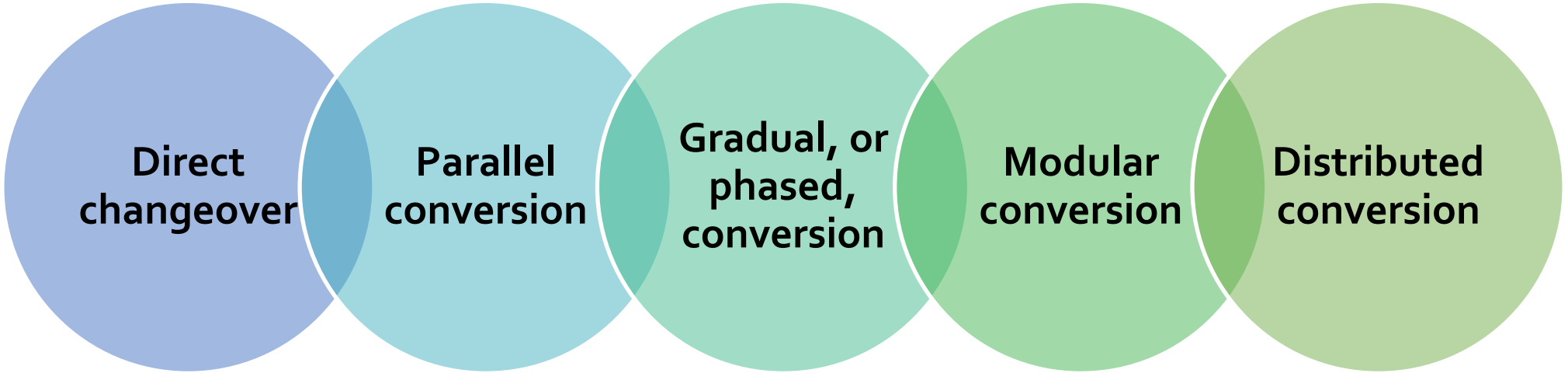
Each user and operator will need slightly different training. To some extent, their jobs determine what they need to know, and their personalities, experience, and backgrounds determine how they learn best. Some users learn best by seeing, others by hearing, and still others by doing. Because it is often not possible to customize training for an individual, a combination of methods is often the best way to proceed. That way, most users are reached through one method or another.

TRAINING STRATEGIES

- Methods for those who learn best by seeing include demonstrations of equipment and exposure to training manuals.
- Those who learn best by hearing will benefit from lectures about procedures, discussions, and question-and-answer sessions among trainers and trainees.
- Those who learn best by doing need hands-on experience with new equipment.

CONVERSION TO A NEW SYSTEM

- A third approach to implementation is physically converting the old information system to the new or modified one. There are many conversion strategies available to analysts.



CONVERSION TO A NEW SYSTEM

Conversion Strategies :

The five strategies for converting from the old system to the new are as follows:

1. **Direct changeover.** (means that, on a specified date, users stop using the old system and the new system is put into use.)
2. **Parallel conversion.** (Parallel conversion refers to running the old system and the new system at the same time, in parallel)
3. **Gradual, or phased, conversion** (Gradual, or phased, conversion attempts to combine the best features of the two previously mentioned plans, without incurring all the risks.)
4. **Modular conversion** (As each module is modified and accepted, it is put into use.)
5. **Distributed conversion** (One entire conversion is done (with any of the four approaches considered previously) at one site. When that conversion is successfully completed, other conversions are done for other sites.)

EVALUATION TECHNIQUES

- Throughout the systems development life cycle, the analyst, management, and users have been evaluating the evolving information systems and networks to give feedback for their eventual improvement.

1. **cost-benefit analysis :**

May be difficult to apply, because information systems provide information about objectives for the first time, making it impossible to compare performance before and after implementation of the system or distributed network.

2. **The revised decision evaluation :**

approach presents difficulty, because all variables involved with the design, development, and implementation of the information system cannot be calculated.

EVALUATION TECHNIQUES

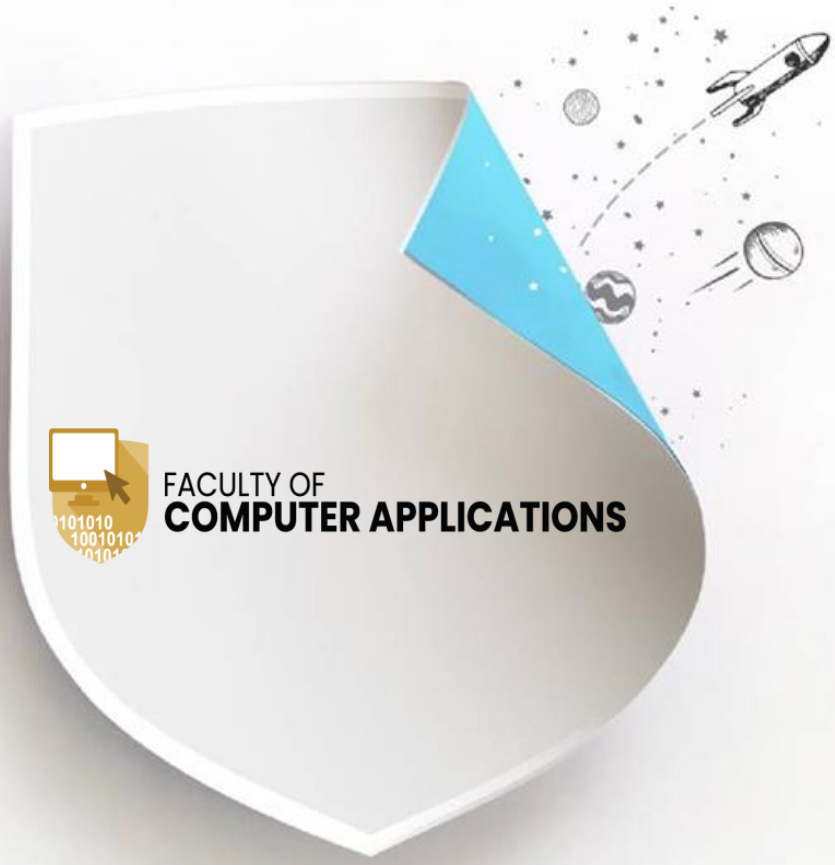
3. **The user involvement approach :**

yields some insight for new projects by providing a checklist of potentially dysfunctional behaviour by various organizational members, but it stresses implementation over other aspects of design.

4. **The information system utility approach :**

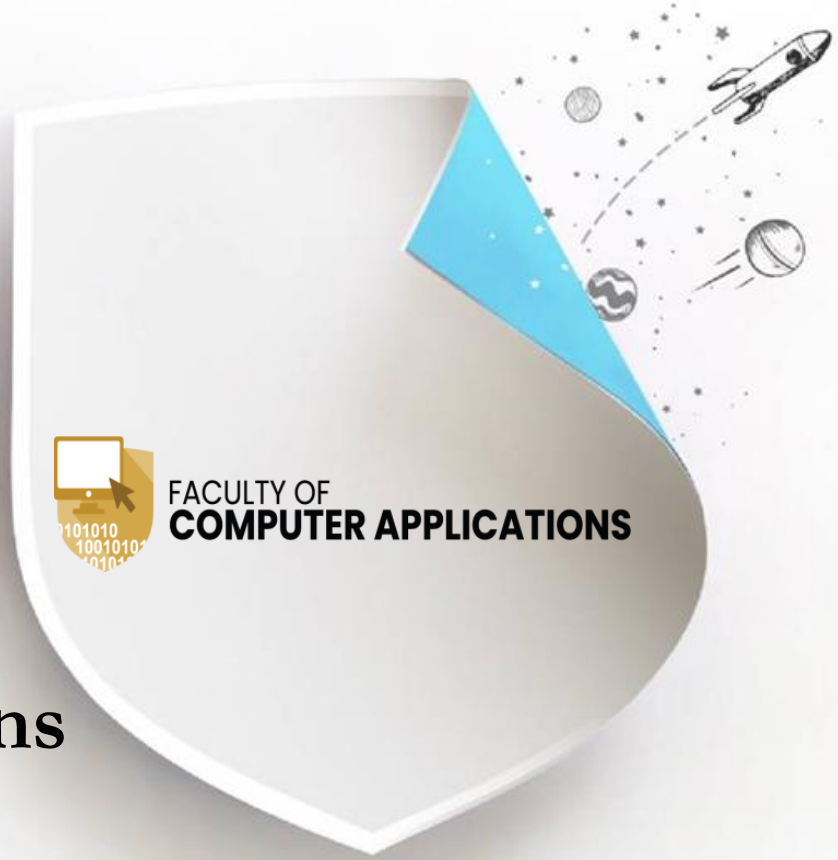
evaluation can be more comprehensive than others if it is expanded and systematically applied.

THANK YOU



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Bachelor of Computer Applications

BCA Semester: 2

05BC3202 – Analysis and Design of Systems